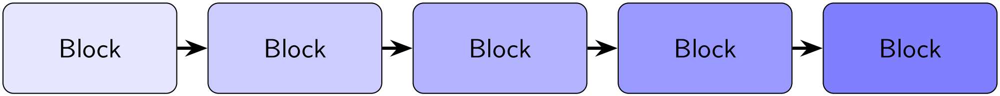
Blockchain 101

Your first smart contract

UMons

What is a blockchain?



A **public database** that anyone can read.
Append-only. Signed. Unstoppable.

Smart contracts = programs that live on it.

The 4 words you need

Wallet	**Contract**	**Transaction**	**Gas**
a key pair $=$ *you*	a program $+$ state	a signed instruction	the price of a write

Reading is **free**. Writing costs **gas**.

Life of a transaction

You **sign** → Network **broadcasts** → Validator **includes** → Block **mined** → State **updated**

Takes a few seconds on Arbitrum. Costs a fraction of a cent.

Once mined, the change is ***permanent*** *and visible to the whole world.*

What we will build

A contract called `SimpleStorage`

`value`

`owner`

`lastValueBy[address]`

`set(uint256 x)`

`increment()`

`reset()` *only owner*

`event ValueChanged(...)`

- Stores a number
- Anyone can change it
- Tracks who wrote what
- Only the deployer can `reset`

What's an event?
A log line the contract emits.
It does *not* change state, but it is **saved on-chain** and anyone can subscribe to it — great for dApps that want to react in real time.

Today's goal

Write a smart contract → **Deploy** on Arbitrum Sepolia → **Call it** 3 different ways

Your code, live on the public Internet.

3 ways to call your contract

1. Explorer	**2. Console**	**3. Script**
click buttons on Arbiscan	REPL on the live chain	TypeScript, the way a dApp does

Same contract. Three angles to understand it.

Read = free. *Write = signed transaction + gas.*

Resources

Docs	`tanguyvans.github.io/blockchain-101`
Repo	`github.com/tanguyvans/blockchain-101`
Network	Arbitrum Sepolia (chain id 421614)
Explorer	`sepolia.arbiscan.io`
Faucet	`faucet.quicknode.com/arbitrum/sepolia`

Need test ETH? Email me your public address at `tanguy.vansnick@umons.ac.be` and I will send you faucet funds.

Let's build.